
SECTION A [Multiple choice questions] - 12 marks

QUESTION 1

The largest interval on which the function $f(x) = \sqrt{3-x} - 5\sqrt{2-x}$ is continuous, is

(a) $(-\infty, 2)$	(b) $(-\infty, 2]$	(c) $(-\infty, 3)$	(d) $(-\infty, 3]$	(e) $(2, 3)$	(f) $[2, 3]$	(g) None of (a) - (f)
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QUESTION 2

$$\lim_{x \rightarrow \infty} 2 \tan^{-1}(2x) =$$

(a) ∞	(b) $\frac{\pi}{2}$	(c) π	(d) 2π	(e) 0	(f) None of (a) - (e)
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QUESTION 3

Consider the following statements:

- (1) If c is in the domain of f and $f'(c)$ does not exist, then c is a critical number of f .
- (2) $x = 0$ is a critical number of the function $f(x) = \ln(x)$.
- (3) $x = 0$ is a critical number of the function $f(x) = \sqrt{x}$.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

ROUGH WORKING

2.1

QUESTION 4

$$\lim_{x \rightarrow -5^+} \frac{x-3}{x+5} =$$

(a) 1	(b) $-\frac{3}{5}$	(c) 0	(d) $-\infty$	(e) ∞	(f) None of (a) - (e)
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4.3

QUESTION 5

Consider the following statements:

- (1) Suppose $f'(c) = 0$. The second derivative test is inconclusive when $f''(c) = 0$.
- (2) Suppose $f'(c) = 0$. The second derivative test fails if $f''(c)$ does not exist.
- (3) When all its tangents lie below the graph of f , the graph is called concave upward.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

4.3

QUESTION 6

Consider the following statements:

- (1) If $f'(x) = g'(x)$ for all $x \in I$, then $f(x) = g(x)$ on I .
- (2) If f has a local extreme value at $x = c$, then $f'(c) = 0$.
- (3) If $f'(c) = 0$, then f has a local extreme value at $x = c$.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

QUESTION 7

Consider the following statements:

- (1) If $f(x) = \ln(3x)$, then $f'(x) = \frac{3}{x}$.
- (2) If $f(x) = \sinh(3x) + \cosh(3x)$, then $f'(x) = 3f(x)$.
- (3) If $f(x) = e^{3x}$, then $f'(x) + f'(-x) = 3(f(x) + f(-x))$.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

QUESTION 8

Consider the following statements:

- (1) $\frac{\infty}{0}$ is an indeterminate form.
- (2) $\lim_{x \rightarrow 0} \frac{x}{e^x} = 1$.
- (3) $\lim_{x \rightarrow -\infty} \frac{x}{e^x} = \infty$.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

End of Section A

SECTION B [Fill in questions] - 13 marks

SECTIONS B AND C MUST BE SCANNED AND UPLOADED ON CLICKUP.

Please do not scan your rough working or any empty pages.

SURNAME:

INITIALS:

STUDENT NUMBER:

SECTION B: Write only your final, fully simplified answer in the block provided. Only this will be marked. Rough working can be done outside the blocks and will not be marked.

QUESTION 9 [1 mark]

If $f(1) = 5$ and $f'(x) \geq 2$ for $1 \leq x \leq 5$, write down the smallest possible value of $f(4)$.

3.2 QUESTION 10 [1 mark]

Suppose $f(2) = 5$, $g(2) = 6$, $f'(2) = 1$ and $g'(2) = -3$. Then $(fg)'(2) =$

1.3 QUESTION 11 [1 mark]

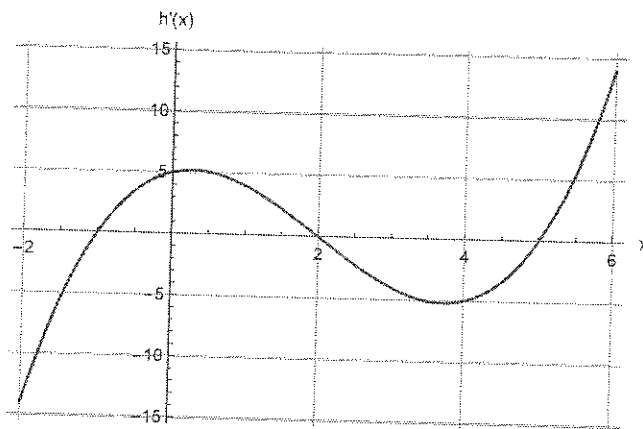
$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} =$$

3.6 QUESTION 12 [1 mark]

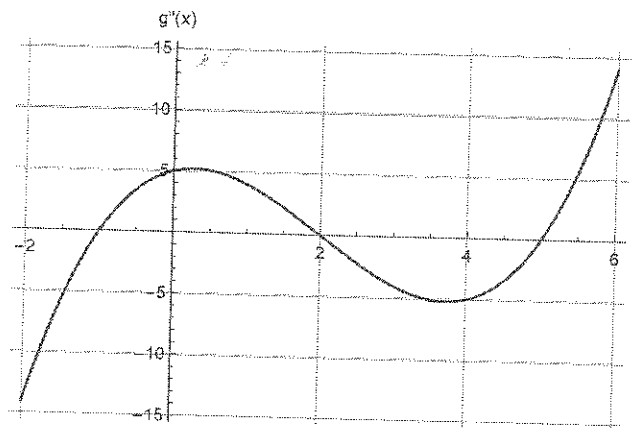
$$\frac{d}{dt}(\ln t)^2 =$$

2.2 QUESTION 13 [1 mark]

The vertical asymptote(s) of the function $f(x) = \frac{x^2 - 5x}{x^3 - 5x^2 + 6x}$ is(are):

4.3
QUESTION 14 [2 marks](a) Consider the graph of h' :Write down the interval(s) on which the function h is increasing.

(1)

(b) Consider the graph of g'' :Write down the interval(s) on which the function g is concave down.

(1)

2.6 QUESTION 15 [2 marks]

The point on the curve $y = \frac{e^{-x}}{x^2 + 1}$ at which the tangent line is horizontal, is

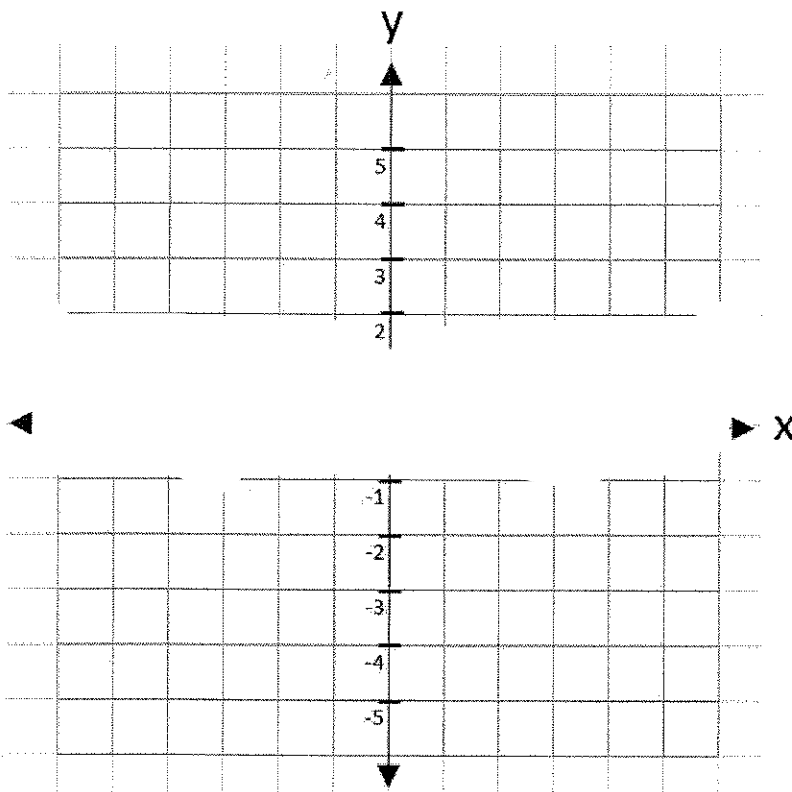
2.3 QUESTION 16 [2 marks]

$$\lim_{\theta \rightarrow 0} \frac{\tan(4\theta)}{4\theta + \sin \theta} =$$

4.1 QUESTION 17 [2 marks]

Sketch the graph of the function f on the axes provided, indicating all relevant intercepts and information, where f is continuous and even and

$$f(0) = 0, f'(x) = 2x \text{ if } 0 < x < 1, f'(x) = -1 \text{ if } 1 < x < 3 \text{ and } f'(x) = 1 \text{ if } x > 3.$$



End of Section B

SECTION C [Full written answers] - 20 marks

2.4 QUESTION 18 [4 marks]

Consider the function $f(x) = \begin{cases} ax + b & \text{if } x < 1 \\ 3 & \text{if } x = 1 \\ bx^2 - a & \text{if } x > 1 \end{cases}$

- 2.4 (a) Find the values of a and b for which $\lim_{x \rightarrow 1} f(x)$ exists. Explain your answer. (2)

- 2.4 (b) Complete the definition:

The function f is continuous at $x = 1$ if

(1)

- 2.4 (c) Find the values of a and b for which the function $y = f(x)$ is continuous at $x = 1$.

(1)

4.4

QUESTION 19 [5 marks]

Determine the limit, if it exist(s).

(a) $\lim_{x \rightarrow -\infty} \sqrt{x^2 - 3x} + x.$ (3)

4.4

(b) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{2\sqrt{x}}.$ (2)

QUESTION 20 [3 marks]

Use the Closed Interval Method to determine the extreme values of $f(x) = \frac{1}{x^2 + x + 1}$ on the interval $[-2, 1]$.

3.5 QUESTION 21 [3 marks]

We know that $\sin^{-1}(x) = y$ if and only if $\sin y = x$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

- (a) Use implicit differentiation and the definition of the function $\sin^{-1}(x)$ to show that

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}. \quad (2)$$

- 3.5 (b) Find $\frac{dy}{dx}$ if $y = \sin^{-1}(\tan x)$.

(1)

QUESTION 22 [3 marks]

4.3 Use the First Derivative Test to determine the local extreme values of the function $f(x) = xe^{2x}$.

QUESTION 23 [2 marks]

4.3 Let f and g be differentiable functions with $f > 0$ and suppose $g = \frac{1}{f}$. Show that g is concave up when f is concave down.